

Adaptive Pac-Man AI - Experiment Report

Map: Cyber Citadel Scenarios: 5 Generated: 2026-09-17T13:54:59.507991+00:00

Algorithm	Success	Avg steps	Avg cost	Avg expanded	Avg frontier	Avg ms
BFS	5/5	41.8	58.2	195.8	11.6	0.3101
DFS	5/5	64.6	90.2	165.8	30.2	0.2954
UCS	5/5	41.8	54.8	194.0	11.6	0.7445
Dijkstra	5/5	41.8	54.8	194.0	11.6	0.5617
A*	5/5	41.8	54.8	151.2	14.6	0.6244

A* heuristic	Success	Avg steps	Avg cost	Avg expanded	Avg frontier	Avg ms
Manhattan	5/5	41.8	54.8	151.2	14.6	0.6021
Euclidean	5/5	41.8	54.8	157.4	16.4	0.6290
Weighted A*	5/5	41.8	54.8	132.6	14.4	0.5522

Interpretation: expanded nodes measure search work; path cost includes terrain and active ghost-danger penalties. Weighted A* may reduce search effort, but unlike Manhattan and Euclidean A*, it does not guarantee an optimal path. Timing is hardware-dependent; compare it together with deterministic steps, cost, frontier, and expanded-node counts.